

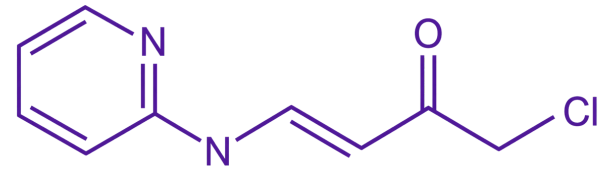
DRUG REPRESENTATIONS

PER-MODALITY ENCODERS

GATED CROSS-ATTENTION FUSION

CLASSIFICATION HEAD

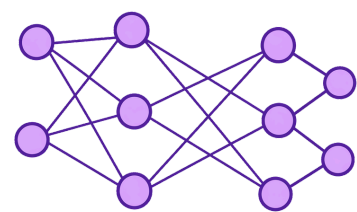
SMILES



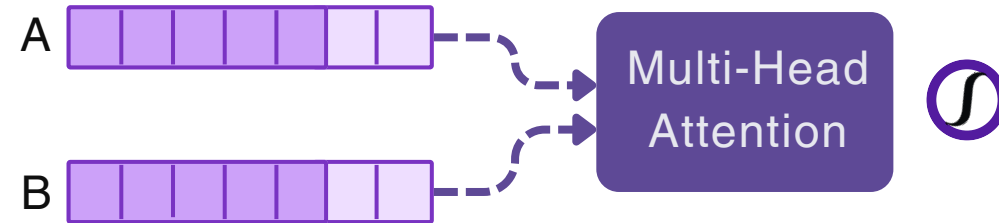
ChemBERTa



CHEMICAL ENCODER



CHEMICAL CROSS-ATTENTION



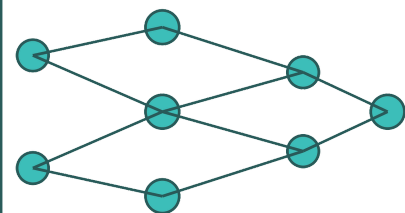
SEQUENCE

MAD
KVLTSKGIK...

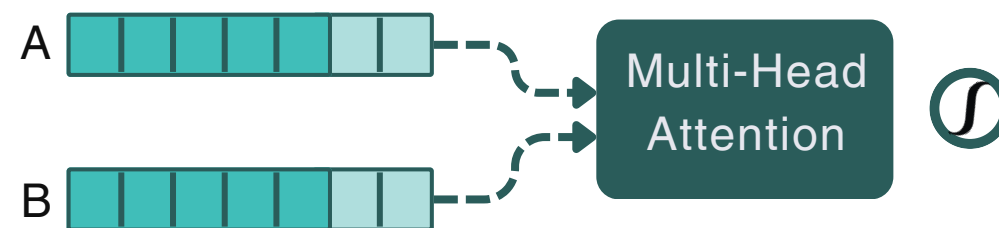
ESM-2



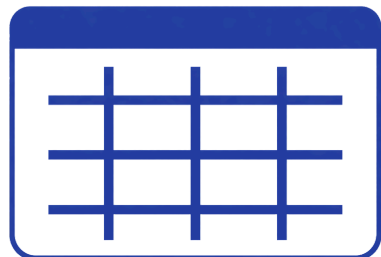
PROTEIN ENCODER



PROTEIN CROSS-ATTENTION



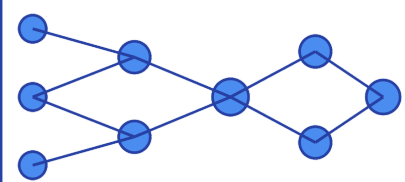
PATHWAYS



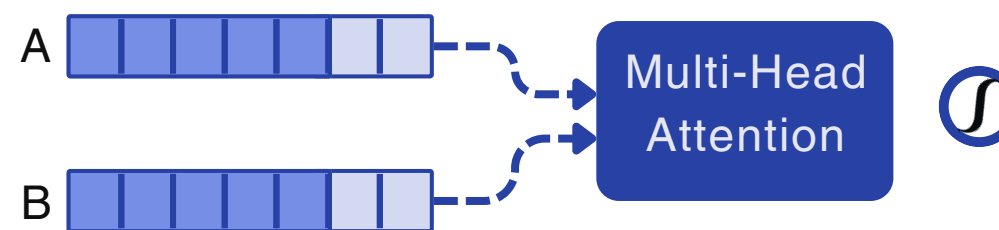
BioBERT



PATHWAYS ENCODER



PATHWAYS CROSS-ATTENTION



MLP

Interaction Probability